

Scanning station: the work cycle — validation and aggregation

This is an informational translation. The matching Russian revision is authoritative. Step-by-step operator guide: scanning codes and station signals, filling and closing boxes, working offline, pausing and closing the shift.

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1. Purpose

This instruction covers the operator's work at a Markiro scanning station during a shift: scanning marking codes, station signals, filling and closing boxes in aggregation mode, working offline and finishing the shift. Signing in and starting a shift are covered by instruction MKR-INS-01.

The shift mode is set when the shift is created: “Validation” — the station only checks and records every code; “Aggregation” — accepted units are additionally packed into boxes, and the station tracks them and prints box labels.

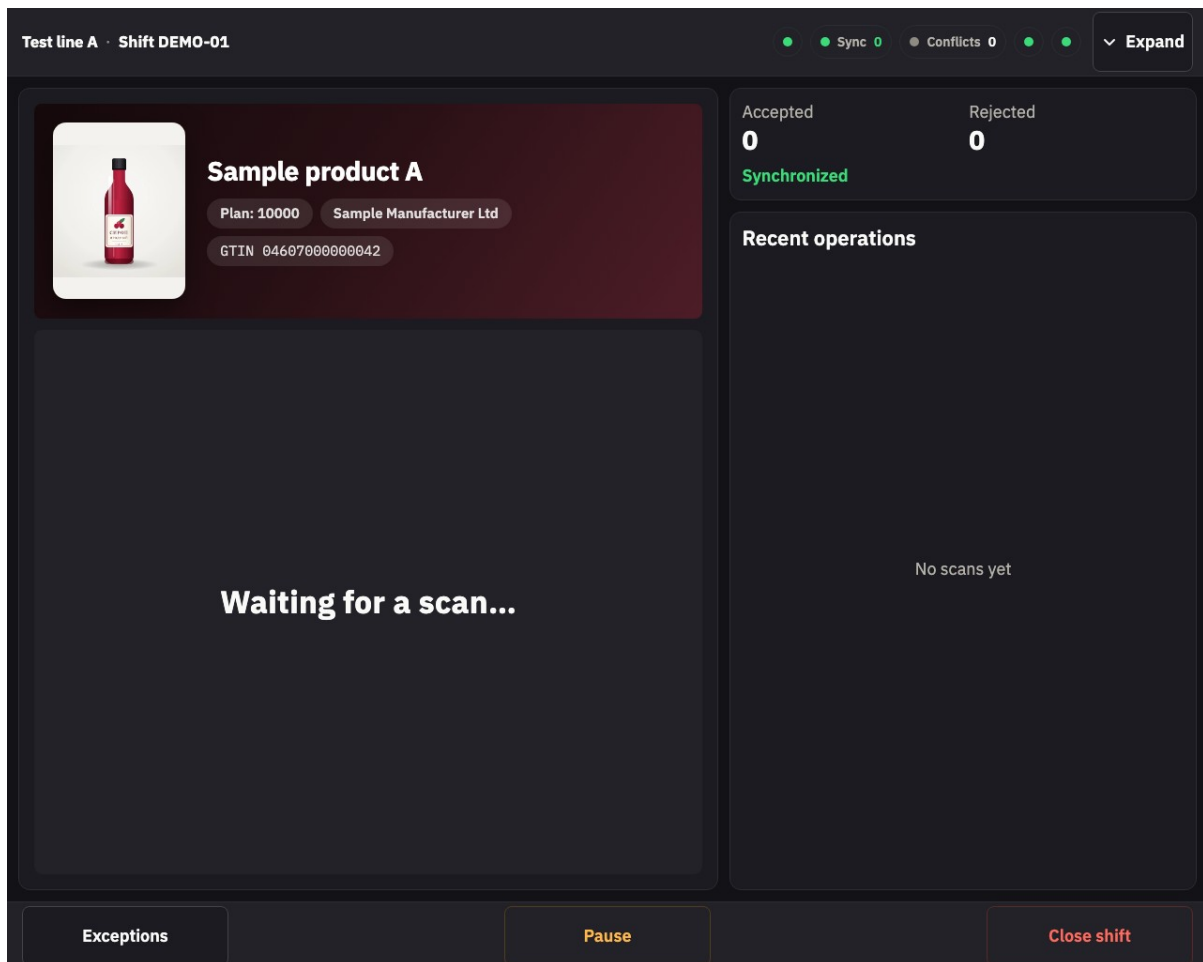
Note: The screenshots use demo data. Product names, shift numbers and counters on your station will differ.

2. The scan cycle and signals

Step 1. Scan a marking code

Take a unit of product and point the scanner at the DataMatrix code. The station processes scans one at a time: wait for the signal for the current unit before scanning the next one.





Work screen: the station is waiting for a scan

Expected result: The station beeped and showed the scan result.

Step 2. Code accepted — keep going

The station shows an accepted scan on the work screen: a green panel with a check mark and the code, and the “Accepted” counter goes up. In aggregation mode, put the unit into the open box; in validation mode, pass it on down the line.



Test line A · Shift DEMO-01

Sync 0

Conflicts 0

Expand



Sample product A

Sample Manufacturer LtdGTIN 04607000000042

(01)04607000000042 (21)DEMO-SERIAL-000128 (91)ABCD (92)TEST-
✓ LONG-CRYPTO-TAIL-ABCDEFGHIJKLMNPOQRSTUVWXYZ-0123456789
(93)XYZ1

Accepted1,248Synchronized

Rejected3

Recent operations

ACCEPTEDGTIN: 04607000000042Serial number: DEMO-SERIAL-0001...2:32:08 PM

ACCEPTEDGTIN: 04607000000042Serial number: DEMO-SERIAL-0001...2:32:07 PM

ACCEPTEDGTIN: 04607000000042Serial number: DEMO-SERIAL-0001...2:32:06 PM

ACCEPTEDGTIN: 04607000000042Serial number: DEMO-SERIAL-0001...2:32:05 PM

ACCEPTEDGTIN: 04607000000042Serial number: DEMO-SERIAL-0001...2:32:04 PM

ACCEPTEDGTIN: 04607000000042Serial number: DEMO-SERIAL-0001...2:32:03 PM

Exceptions

Pause

Close shift

Work screen: the code is accepted (green panel)

Step 3. The “DUPLICATE” signal – set the unit aside

This code has already been scanned: the station shows the time of the first scan. Do not put the unit into a box — set it aside separately. If duplicates keep coming, stop and tell your supervisor: the product may have already passed through the station.



**DUPLICATE**

First scanned at 2:31:52 PM

The “DUPLICATE” signal with the time of the first scan

Step 4. A red signal — do not let the unit through

“WRONG CODE” — the scan was not a marking code or the code is damaged: flatten the packaging and scan again; if the code cannot be read, set the unit aside. “WRONG GTIN” — the product does not belong to this shift: remove it from the line. “WRITE FAILED” — the station could not save the scan: stop and call a technician.





A red signal: the unit must not pass

Note: The full-screen signal fades out by itself after a few seconds. If you got distracted, check the “Recent operations” panel: it lists the latest scans with verdicts and times.

3. “Validation” mode

In validation mode the operator's job is to run every unit through the scanner and watch the signals. There are no boxes and no printing in this mode.

- The “Accepted” and “Rejected” counters show the shift progress; if a plan is set, “Plan: N” is shown under the product.
- The “Recent operations” panel shows recent scans with serial numbers and times.
- When the plan is reached, the station reports “Plan completed” — follow your supervisor's directions from there.

4. “Aggregation” mode

Step 1. Fill the open box

The box panel on the work screen shows its number (“Box no. 1”) and the item count. Put a unit into the box only after the green accepted-code panel.





Test line A · Shift DEMO-01

Sync 0

Conflicts 0

Expand



Sample product A

Sample Manufacturer Ltd

GTIN 04607000000042

Box no. 1

2 / 10 Items

✓ DEMO-SERIAL-000128

1

2

3

4

5

6

7

8

9

10

Close box

Undo last scan

Clear box

Accepted

Rejected

1,248

3

Synchronized

Recent operations

ACCEPTED

GTIN: 04607000000042

Serial number: DEMO-SERIAL-0001...

2:32:08 PM

ACCEPTED

GTIN: 04607000000042

Serial number: DEMO-SERIAL-0001...

2:32:07 PM

ACCEPTED

GTIN: 04607000000042

Serial number: DEMO-SERIAL-0001...

2:32:06 PM

ACCEPTED

GTIN: 04607000000042

Serial number: DEMO-SERIAL-0001...

2:32:05 PM

ACCEPTED

GTIN: 04607000000042

Serial number: DEMO-SERIAL-0001...

2:32:04 PM

ACCEPTED

GTIN: 04607000000042

Serial number: DEMO-SERIAL-0001...

2:32:03 PM

Exceptions

Pause

Close shift

Aggregation work screen: the open box panel

Step 2. Close the full box

When the box reaches its capacity, close it with the “Close box” button. The station assigns the box an SSCC number and sends the label to the printer. Stick the label onto this box right away — do not put it aside.



[illegible]

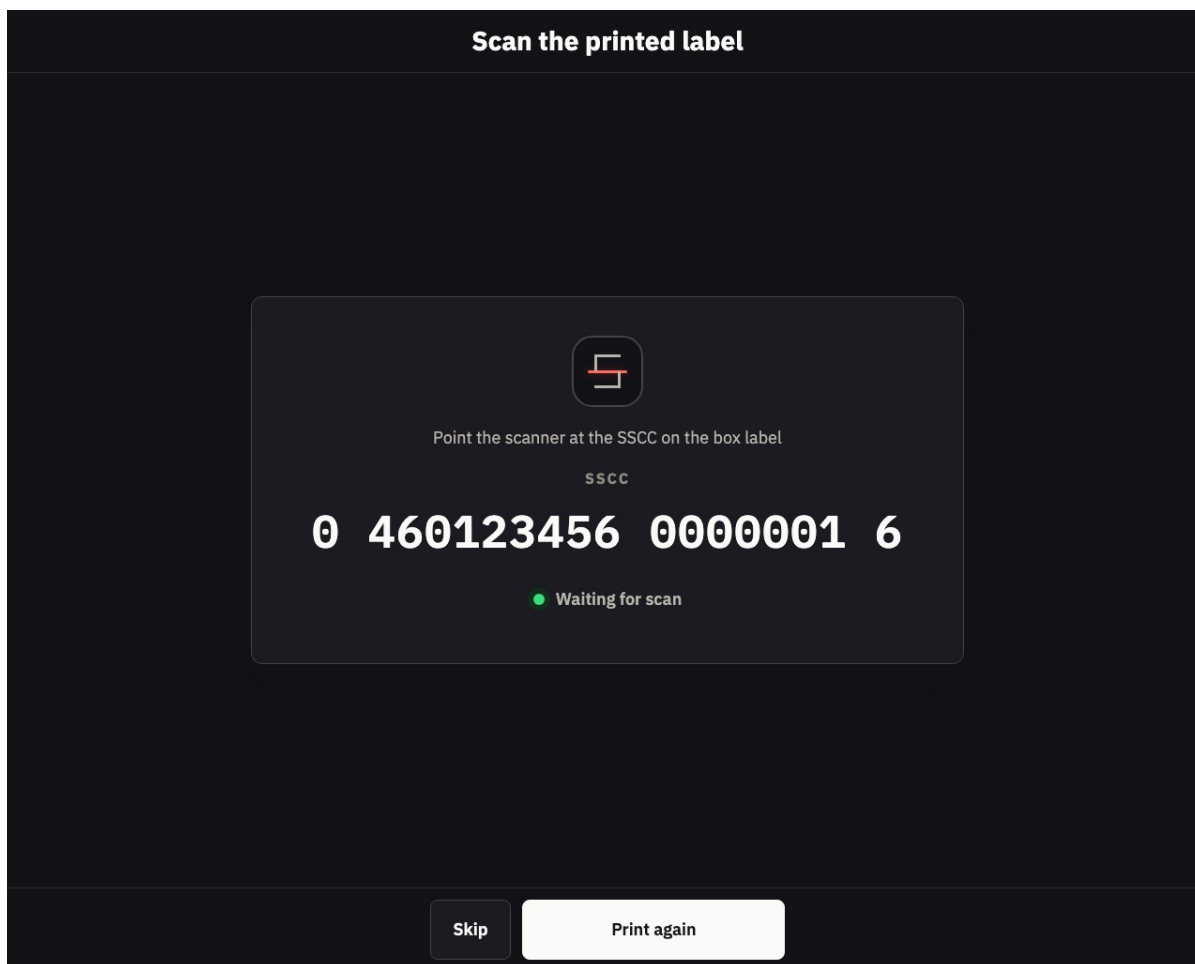
The box is full and ready to be closed

Expected result: The printer printed the box label.

Step 3. Verify the printed label

If print verification is enabled on the station, the “Scan the printed label” screen appears: point the scanner at the SSCC code on the label. The “This is a different label” message means you are holding a label from another box — find the right one or tap “Print again”. The “Skip” button skips the verification; the skip is recorded.





Verifying the printed box label

Expected result: The station confirmed the label and opened the next box.

Important: Every label belongs to one specific box: the SSCC number is unique. A label stuck onto the wrong box breaks the records of the whole batch.

Note: A partially filled box at the end of the shift is closed with the same “Close box” button. “Undo last scan” removes the last added item from the box, and “Clear box” removes every item from the open box — use them only when your supervisor says so.

5. Working offline

Step 1. Keep working during “No connection”

If the status bar shows “Server: No connection”, the station keeps accepting scans and stores them locally — the “pending” counter shows the queue. Work as usual: once the connection is back, the data goes to the server by itself.





Test line A · Shift DEMO-01

● Server No connection ● Sync 7 ● Conflicts 0 ● Expand



Sample product A

Sample Manufacturer LtdGTIN 04607000000042

(01)04607000000042 (21)DEMO-SERIAL-000128 (91)ABCD (92)TEST-
✓ LONG-CRYPTO-TAIL-ABCDEFGHIJKLMNOSTUVWXYZ-0123456789
(93)XYZ1

Accepted
1,248
7 pending

Rejected
3

Recent operations

ACCEPTEDGTIN: 04607000000042
Serial number: DEMO-SERIAL-0001...2:32:08 PM

ACCEPTEDGTIN: 04607000000042
Serial number: DEMO-SERIAL-0001...2:32:07 PM

ACCEPTEDGTIN: 04607000000042
Serial number: DEMO-SERIAL-0001...2:32:06 PM

ACCEPTEDGTIN: 04607000000042
Serial number: DEMO-SERIAL-0001...2:32:05 PM

ACCEPTEDGTIN: 04607000000042
Serial number: DEMO-SERIAL-0001...2:32:04 PM

ACCEPTEDGTIN: 04607000000042
Serial number: DEMO-SERIAL-0001...2:32:03 PM

Exceptions

Pause

Close shift

Working offline: scans accumulate on the station

Expected result: After the connection is restored the counter goes down to “Synchronized”.

Important: Call an administrator if the “pending” counter does not go down for a long time while the network is up, or the status bar shows “Sync: Not syncing”. In aggregation mode a long offline period can exhaust the box number reserve — the station will report “Box numbers have run out” and pause scanning until the connection is back.

6. Pausing and closing the shift

Step 1. Take a break through “Pause / finish”

The “Pause” button suspends work for a break. “Leave shift” releases the station without closing the shift — you or another operator can continue it after the break. If some scans have not reached the server yet, the station warns about it; the data stays on the station and is sent once there is a connection.

Step 2. Close the shift at the end of work

Tap “Close shift”. In aggregation mode close the open box first — the station reminds you: “Close the open box first”. If the actual quantity does not match the plan, the station asks for the reason for the difference. After closing, the “Shift summary” is shown.

Expected result: The shift is closed and the station returned to the shift selection screen.

7. Common problems

Duplicates keep coming one after another — This product has probably been scanned before. Stop, set the batch aside and call your supervisor.





The printer did not print the box label — The station shows the “The box label was not printed” screen with options to reprint or continue without a label. Print failure handling is covered in detail by the exceptions instruction.

Box numbers have run out — The SSCC number reserve was exhausted while offline. Restore the station's connection (call an administrator), then get back to work.

The “pending” counter grows although the network is up — Do not stop working — scans are not lost. Tell an administrator: the send queue needs attention.

The scanner stopped reading codes — Check the scanner indicator in the status bar — a problem indicator expands into a caption, and the “Expand” button opens the full view — and check the cable. If the connection does not come back, call a technician.

If your problem is not listed above, contact your organization's administrator or Markiro support: hello@v-b.tech.

